SOFTWARE ENGINEERING QUIZ

Topic: Requirements Engineering and Writing Requirements

Time Allowed	50 minutes	**Total Marks**	30
Mode	Open-book / Internet allowed	**Submission**	Handwritten

Instructions:

This quiz is to be completed **by hand on paper**.

- Write your **Name, Registration Number, Section, and Date** on the first page.
- Read all questions carefully before starting.
- Answer **all questions** in your own handwriting.
- Keep your answers **clear, concise, and relevant to the CEBS case study**.
- For rewritten requirements, use proper requirements language such as **"shall"** where appropriate.
- Show all working for any calculation or ranking question.
- Number each answer clearly according to the question paper.
- After finishing, **scan or take clear pictures** of your handwritten pages and upload them as **one PDF file**.
- Make sure all pages are in the correct order and clearly readable.
- File name format: **RegNo_Name_Quiz.pdf**
- Late submissions may have marks deducted according to course policy.
- Illegible handwriting may result in loss of marks if the answer cannot be read.

CASE STUDY: CAMPUS EQUIPMENT BOOKING SYSTEM (CEBS)

A university plans to develop a Campus Equipment Booking System (CEBS) for laptops, projectors, cameras, microphones, and tablets. Students should be able to search available equipment, submit booking requests, and view booking status online. Lab technicians inspect returned items, mark damaged items as unavailable, and confirm actual return times. Faculty members may approve extension requests for selected equipment. The administration wants monthly utilization reports and better visibility of which items are available, reserved, damaged, or under maintenance. High-value items such as cameras

and laptops must require identity verification at pickup. Equipment may be unavailable because of prior reservations, maintenance, or damage.

Question 1 — Classification and stakeholder understanding (4 marks)

For each statement, identify whether it is **FR**, **NFR**, or **Domain Requirement**. Then name the most relevant stakeholder and justify your answer in one brief line.

ID	Statement	Type	Stakeholder + 1-line justification
S1	The system shall allow students to search equipment by category and current availability.	*FR / NFR / Domain*	
S2	Search results shall appear within 3 seconds on the university Wi-Fi network under normal load.	*FR / NFR / Domain*	
S3	Pickup of cameras and laptops shall require a valid university ID.	*FR / NFR / Domain*	
S4	Borrowing records shall be retained for 1 year to satisfy university audit policy.	*FR / NFR / Domain*	

Question 2 — Review and rewrite flawed requirements (12 marks)

For each requirement: **(a)** identify the **main** quality problem, and **(b)** rewrite it as a better requirement. Choose the main issue from terms such as ambiguous, not testable, not atomic, implementation-free violated, incomplete, not independent, or unrealistic.

Given requirement	Main issue	Improved requirement
R1. The system shall quickly show available equipment to users.		*Rewrite using 'shall' and measurable wording*
R2. The system shall store reservation data in Firebase for better performance.		*Rewrite using 'shall' and measurable wording*
R3. A student shall be able to reserve equipment, cancel the reservation, and request an extension.		*Rewrite using 'shall' and measurable wording*
R4. The system shall display an estimated pickup time for every request.		*Rewrite using 'shall' and measurable wording*

Reminder for Question 2: A strong rewrite should be clear, specific, testable, and free from unnecessary design choices. When a requirement combines multiple actions, split it into separate requirements.

Question 3 — Validation and acceptance tests (6 marks)

Choose any two of your rewritten requirements from Question 2. For each one, write **one validation check** and **one concrete acceptance test condition**. Keep them specific to the CEBS case.

Requirement ID	Validation check / review question	Acceptance test condition
e.g., R1		
e.g., R1		

Tip: Use checks that connect to validation, realism, verifiability, or consistency. Your acceptance test should describe a clear condition and expected result.

Question 4 — Prioritization and change impact (8 marks)

Part A (5 marks): Complete the prioritization work below and rank the requirements from highest to lowest priority. Use the class idea that **higher benefit and penalty increase value**, while **higher cost and risk reduce priority**.

Quick formula reminder: Value Total = (2 × Benefit) + (0.5 × Penalty). You may use brief calculations and then state the final ranking.

Candidate requirements:

- P1. Search equipment by category and date — B=8, P=6, C=3, R=2
- P2. University ID verification at pickup for high-value items — B=7, P=8, C=2, R=2
- P3. Monthly utilization reports for admin — B=5, P=4, C=2, R=1
- P4. Return reminder notifications 2 hours before due time — B=6, P=5, C=3, R=1

Req	B	P	Value Total	C	R	Rank
P1	8	6	*calc*	3	2	*1-4*
P2	7	8	*calc*	2	2	*1-4*
P3	5	4	*calc*	2	1	*1-4*
P4	6	5	*calc*	3	1	*1-4*

Write your final ranking here (highest to lowest): _____

Part B (3 marks): A new change request arrives — **"guest researchers should also be allowed to borrow equipment."** State **two impacts** on existing requirements and **one requirements management action** that the team should take.